Solution Architecture

Maximum Marks: 3 Marks

Date	Team ID	Project Name
15 March 2026	Batch 2023-27 | ACOE	Rising Waters – ML-Based Flood Prediction System

Team Members

S.No	Team Member Name	Role
1	Venkatesh Balireddy	Team Lead
2	Archana Dhanani	Member
3	Shivatmika Gandikota	Member
4	Gode Siva Ramakrishna Durgaprasad	Member
5	Bavisetty Gopi Krishna	Member

5-Layer System Architecture

The Rising Waters system follows a layered architecture separating user interaction, presentation, application logic, machine learning, and data storage concerns.

Layer	Layer Name	Components	Technology
1	User Layer	Web Browser (Chrome, Firefox, Edge)	HTML5 / CSS3 / JS
2	Presentation Layer	Home, Predict, Flood Result, No Flood Result, History, About pages	Jinja2 Templates
3	Application Layer	Flask Routes, Form Validation, Session Handling, Prediction Logging	Python 3.11 + Flask 2.x
4	ML Layer	XGBoost Model (floods.save), StandardScaler (scaler.save), Joblib	Scikit-learn + XGBoost
5	Data Layer	flood_dataset.csv (training data), predictions_log.csv (audit log)	Pandas / CSV
6	Deployment Layer	Docker Container → IBM Cloud Foundry (manifest.yml + Procfile)	Docker + IBM Cloud

Request Flow

Step	Action	Component
1	User opens browser and navigates to / (Home)	Browser → Flask GET /
2	User fills in 6-field form and submits	Browser → Flask POST /predict
3	Flask validates inputs; returns error if invalid	utils/preprocessing.py
4	Valid inputs scaled using pre-fitted StandardScaler	scaler.save (Joblib)
5	Scaled array passed to XGBoost model for inference	floods.save (XGBoost)
6	Result label (0/1) and probability stored in Flask session	app.py session
7	Flask redirects to /result/flood or /result/noflood	result_flood.html / result_noflood.html
8	Prediction logged to predictions_log.csv with timestamp	CSV file append